

Sentinel Project Proposal

Executive Summary

Sentinel is a life-saving mobile application powered by NASA Earth observation data and local community input. It provides real-time alerts, risk mapping, and actionable guidance tailored to each user's location. Whether it's a conflict zone, natural disaster, or urban emergency, Sentinel adapts to show safe routes, nearby shelters, and critical instructions. Unlike traditional systems, Sentinel works **offline**, ensuring accessibility even when infrastructure is disrupted. By merging satellite insights, predictive modeling, and crowdsourced reporting, it empowers citizens and urban planners to make faster, safer, and smarter decisions — protecting millions of lives worldwide.

Problem Definition

Over 71.1 million people were displaced in 2022 due to conflicts and natural disasters — a 20% increase from the previous year. Emergency response systems are often underprepared, fragmented, or inaccessible during crises. The absence of reliable communication, especially in low-connectivity areas, leaves millions vulnerable and without timely support. Sentinel addresses this gap directly.

Background & Literature Review

Existing emergency alert systems and disaster apps often rely heavily on internet access and centralized reporting. While solutions like early-warning systems exist for specific disasters (e.g., earthquakes or floods), there is no unified, adaptable tool that combines satellite data, predictive modeling, and crowdsourced local reporting into one accessible platform. Sentinel fills this critical gap by integrating NASA Earth observation data with on-the-ground user inputs, ensuring relevance across diverse crises.

Methodology

1. Collect NASA Earth observation data (weather, land use, disaster hotspots).
2. Process data with predictive algorithms to forecast risks (natural disasters, unsafe zones).
3. Integrate crowdsourced inputs for real-time situational awareness.
4. Enable offline functionality to store and broadcast critical alerts.
5. Provide tailored guidance per location (e.g., shelters, safe routes, or protection steps).

Solution

Sentinel adapts to each location: guiding users in war zones to safe shelters, helping citizens in flood-prone regions find evacuation routes, and providing urban communities with air quality alerts or safe public health recommendations. Its offline-first design ensures continuity even when communications collapse.

Value Proposition

Sentinel uniquely combines global satellite intelligence with hyper-local input, bridging the gap between technology and human need. Unlike single-purpose apps, it is versatile, scalable, and focused on saving lives across diverse crises worldwide.

Roles of Team Members

- Project Lead (Vision & Coordination)
- Backend Developer (Data & Algorithms)
- Frontend Developer (Mobile Interface)
- Data Scientist / GIS Specialist (Earth data processing)
- UI/UX Designer (User-friendly design)
- Communications Lead (Pitch & Storytelling)

Workflow Strategy

Our workflow includes problem analysis, data integration, prototype development, and user testing within the hackathon timeframe. We will collaborate in sprints, with checkpoints every 12 hours, to

ensure iterative progress and alignment.

Resources

- NASA Earth observation datasets
- Open-source disaster prediction algorithms
- GIS and mapping tools
- Mobile development frameworks
- Reference: Internal Displacement Monitoring Centre (IDMC) 2023 Report

Team Name: Sentinel

Members:

- 1 - Project Lead
- 2 - Backend Developer
- 3 - Frontend Developer
- 4 - Data Scientist
- 5 - UI/UX Designer
- 6 - Communications Lead